

POMI: A Corpus to Support a Natural Language Inference-based Approach for Detecting Misinformation in Philippine Online News

ABSTRACT & CORE RESEARCH CONTRIBUTIONS

Aimed at combating misinformation through Artificial Intelligence-assisted fact-checking, this paper introduces the Philippine Online Misinformation Inference (POMI) dataset, a compilation of 10,132 factual and false claims from Philippine fact-check articles, and the first misinformation corpus in the Philippines utilizing Natural Language Inference (NLI).

BENCHMARKING EXPERIMENTS & PERFORMANCE DYNAMICS:

In zero-shot and fine-tuned evaluations, the authors benchmarked leading large and lightweight language models:

? LLaMA 3.1 70B achieved zero-shot accuracy of 85.1%, outperforming DeepSeek-R1 8B, LLaMA 3.3 70B, and DeepSeek-R1 70B.

? Fine-tuning lightweight open-weight models produced drastic gains: LLaMA 3.1 8B accuracy escalated from 78.7% to 94.5%.

? This empirical result proves the viability of lightweight, compute-efficient local transformers for Philippine NLI tasks.

DATASET ARCHITECTURE & AVAILABILITY:

? Scope: 10,132 claims curated from verified Philippine fact-checking consortiums (Tsek.ph, VERA Files, Rappler).

? Annotation Scheme: 3-class directional NLI formulation (Entailment, Contradiction, Neutral).

? Domain: Philippine English, Tagalog, and colloquial code-switched Taglish news headlines.

? Public Corpus Link: Google Sheets repository (10,132 annotated pairs)

? Annotation Guidelines: Google Drive materials repository

CITATION & RELEVANCE TO LOCAL ORDINANCE CONFLICT DETECTION:

Núñez et al. (2026) directly validates that transformer architectures can resolve directional logical contradictions in Philippine low-resource and code-switched environments. In our Ateneo de Davao thesis, this foundational NLI framing is adapted and extended from news misinformation triage to ex-ante statutory preemption conflict detection between proposed Davao City ordinances and superior national statutes.